Polina Baron

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Academic Appointments

Postdoctoral Fellow | University of Toronto (St. George), Department of Mathematics (ON, Canada) 2026–present
Supervisor: Kasra Rafi.

Education

University of Chicago, Department of Mathematics (Chicago, IL, USA) Ph.D. 2020–2026
Advisor: Simion Filip. M.Sc. 2020–2022
Dissertation: "Semisimplicity, Rigidity, and Unique Ergodicity in Products and Branched Covers of Translation Surfaces".
Amick Fellowship.

Higher School of Economics, Department of Mathematics (Moscow, Russia) M.Sc. 2020–2022
Advisor: Sasha Skripchenko B.Sc. 2016–2020
Thesis: "Ergodic Properties of Linear Involutions".
Academic scholarship for excellent students. Graduated *Summa Cum Laude* ("s otlichiem").

Research Profile

My research is in dynamical systems and ergodic theory, with an emphasis on Teichmüller dynamics and translation surfaces. I study Hodge-theoretic rigidity of the Kontsevich–Zorich cocycle, unique ergodicity of branched covers, and rigidity phenomena for infinite-type translation surfaces and interval exchange transformations. Additionally, I work on integrable systems, particularly the KdV hierarchy and its algebraic-geometric solutions. I am also interested in connections between translation surfaces and quantum chaos.

Publications and Preprints

P. Baron. **Semisimplicity and rigidity of the Kontsevich–Zorich cocycle for products of strata**. arXiv:2610.21247, 2026.

Establishes structural rigidity results for the Kontsevich–Zorich cocycle over products of strata of translation surfaces.

P. Baron and E. Shuvaeva. **Unique ergodicity of branched covers of translation surfaces**. arXiv:2606.05492, 2026.

Resolves a conjecture of Jon Chaika concerning the preservation of unique ergodicity under branched coverings, establishing new criteria for statistical regularity.

P. Baron. **The Neumann–Moser dynamical system and the Korteweg–de Vries hierarchy**. arXiv:2402.18079, 2024.

Connects the Neumann–Moser system with the KdV hierarchy, relating finite-dimensional integrable dynamics to nonlinear differential equations.

P. G. Baron. **The Mumford dynamical system and the Gelfand–Dikii recursion**. *Functional Analysis and Its Applications* **57** (2023), no. 4, 279–287.

Explores the relationship between the Mumford integrable system and the Gelfand–Dikii recursion, connecting algebraic geometry and integrable dynamics.

Selected Talks and Presentations

Fields Institute. Seminar talk. Toronto, ON (Canada). September 23, 2026

Institute for Advanced Study. Invited seminar talk. Princeton, NJ (USA). May 6, 2026

Dynamics Student Days. Conference talk. CUNY Graduate Center, New York, NY (USA). May 4–6, 2026

University of Chicago. Seminar talk. Chicago, IL (USA). February 16, 2026

Northwestern University. Invited seminar talk. Evanston, IL (USA). January 27, 2026

Joint Mathematics Meetings. Washington, DC (USA). January 4–7, 2026

- AWM Special Session on Research Developments by Women in Mathematical Physics. Invited conference talk.
- AMS Session on Geometric, Dynamical and Probabilistic Aspects of Group Theory. Conference talk.
- AMS Contributed Paper Session on Dynamical Systems and Ergodic Theory. Short talk.

Stony Brook University. Invited seminar talk. Stony Brook, NY (USA). November 21, 2025

Semi-Annual Workshop on Dynamical Systems and Related Topics. Short talk. Pennsylvania State University (USA). November 13–16, 2025

FRG Fall School on Quantization and Lagrangians. Poster presentation. University of Illinois Chicago (USA). September 6–7, 2025

Beyond Hyperbolicity. Poster presentation. ICTP, Trieste (Italy) June 9–13, 2025

Semi-Annual Workshop on Dynamical Systems and Related Topics. Short talk. University of Maryland (USA). April 4–7, 2024

Équations différentielles motiviques et au-delà. Invited online talk. Institut Henri Poincaré (France). March 18, 2024

School on Flat Surfaces and Interactions. Short talk. University of Bordeaux (France). March 4–8, 2024

Additional Research Experience

Research Intern | Microsoft Research, Machine Learning and AI group (New York, NY) Summer 2024
Worked on an "astral space" approach to convex optimization at infinity. Studied convergence of sets and convex functions and the behavior of gradient and stochastic-gradient methods when minimizers escape to infinity. Mentors: Miroslav Dudík and Robert E. Schapire.

Research Intern | Institute for Mathematical and Statistical Innovation (Chicago, IL) Summer 2023
Served as the project's primary mathematical-methods consultant. Developed and applied statistical physics, persistent homology and topological data analysis methods to study criticality in deterministic and probabilistic simulations of functional brain connectivity for Alzheimer's disease research. Mentor: Alex Leow.

Junior Research Assistant | Higher School of Economics (Moscow, Russia) 2018–2021
Studied the ergodic properties of foliations, flows, and linear involutions on surfaces. Supported by joint RFBR–CNRS grant #18–51–15010.

Teaching Leadership and Academic Service

Organizer, Dynamics Pre-Seminar | University of Chicago 2025–2026
Co-organized a regular graduate pre-seminar in dynamical systems, including selecting topics and coordinating speakers.

Phoenix STEM Collaborative Learning Coordinator | University of Chicago 2025–2026
Coordinated 50% of the tutorial sections for the multi-section MATH 131–132 calculus sequence (approximately 400 students in total). Observed tutorial instructors and provided pedagogical feedback, co-developed tutorial materials, and completed accompanying training in collaborative-learning pedagogy.

Organizer, Warm-Up Program for New Graduate Students | University of Chicago September 2024
Organized a week-long academic orientation program for incoming mathematics graduate students, including scheduling sessions and coordinating presenters.

Senior Mentor and Tutorial Coordinator | University of Chicago Mathematics REU Summer 2022
Mentored undergraduate participants on research and exposition and supported the preparation of their final presentations. Coordinated the REU tutorial program, including scheduling tutorials, developing materials, and supporting tutorial leaders.

Teaching Experience

Course Instructor | University of Toronto
MAT136H1S: Calculus II Winter 2027 (scheduled)
Appointed to teach a lecture section within a coordinated, multi-section course.

Course Instructor | University of Toronto
MAT244H1F: Introduction to Ordinary Differential Equations Fall 2026
Independently responsible for a lecture section of approximately 200 students. Designed and delivered all lectures; prepared original lecture notes and course materials for each class; held office hours; and contributed to assessment design and course administration.

Course Instructor | University of Chicago
MATH 112: Studies in Mathematics I Fall 2024
Independently responsible for a class of approximately 20 students. Designed and taught an introduction to number theory, including lectures, course materials, assignments, and assessments.

Course Instructor | University of Chicago
MATH 151–153: Calculus I–III 2023–2024
Independently led the calculus sequence for a class of about 30 students. Prepared lectures and course materials and was responsible for assessment, office hours, and grading.

Lecturer | University of Chicago
MATH 131–133: Elementary Functions and Calculus I–III 2022–2023
Taught lecture section of approximately 30 students throughout a coordinated, multi-section, three-quarter calculus sequence. Prepared and delivered lectures in accordance with the shared syllabus and course policies; held office hours; administered and graded common course assessments.

Teaching Assistant | University of Chicago 2021–2022
Courses included Real Analysis I, Inquiry-Based Learning in Real Analysis II, Ordinary Differential Equations, and Brownian Motion. Led problem-solving sessions, held office hours, and graded coursework.

Skills, Languages, and Interests

Computer & Data Science: computational dynamics, convex optimization, statistical machine learning, topological data analysis.

Technical: Python (incl. Matplotlib, NumPy, scikit-learn, SciPy), SageMath, Wolfram Mathematica, LaTeX.

Languages: English (fluent: IELTS 8.0, GRE Writing 4.5), French (technical reading), Russian (native).

Interests: Creative writing (1st place in a contest of speculative fiction short stories in Russian with 500 participants).